

napatech

HOT INTERCONNECTS 2026

AI Fabrics in Motion

Programmable Interconnects for
Scale-Up, Scale-Out, and Scale-Across

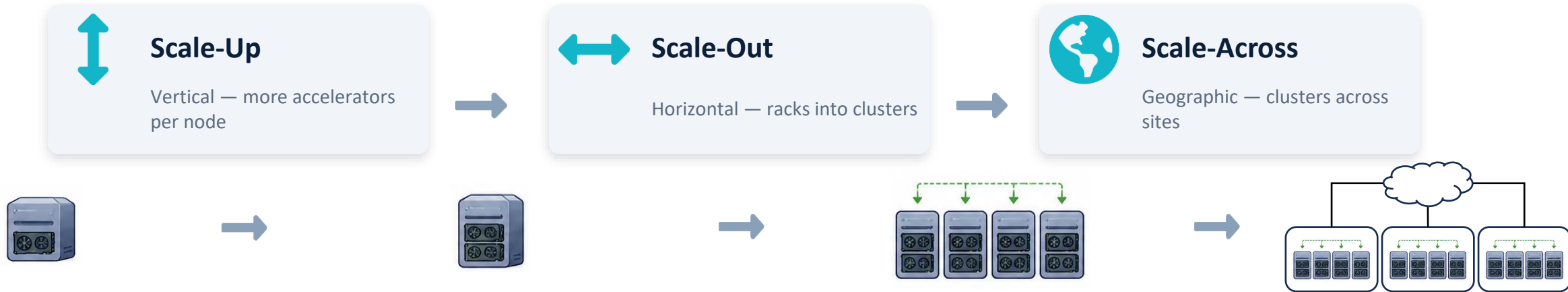
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The bottleneck has moved

As AI clusters scale up, then out, then across, the fabric — not the accelerator — increasingly sets the ceiling on system performance.



● Training fabrics

- Collective-communication overhead (AllReduce)
- Congestion & hotspot formation on shared paths
- Synchronization barriers stall the whole job
- GPU idle time waiting on the network

● Inference fabrics

- KV-cache and token-routing traffic
- Tail latency in distributed serving
- Resource locality & placement sensitivity
- Cross-cluster coordination at scale

TAKEAWAY As AI systems scale, networking adaptability and efficiency increasingly determine overall system performance.

Innovation is outrunning the network

AI models, frameworks, and infrastructure software now evolve faster than fixed-function silicon can be re-spun — opening a gap across four domains at once.



Transport

- Enhanced RoCE
- UEC transport evolution
- Congestion-control innovation



Traffic Engineering

- Flow spraying
- Adaptive routing
- Dynamic load balancing



Collective Communications

- AllReduce optimization
- In-network aggregation
- Hierarchical collectives



Observability

- Real-time telemetry
- Congestion visibility
- Workload-aware monitoring

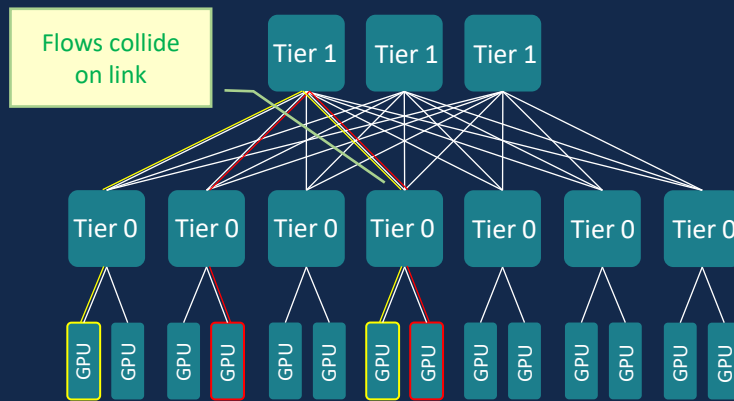
TAKEAWAY *AI infrastructure requirements are evolving faster than traditional network deployment cycles.*

THE ANSWER

Make the fabric programmable

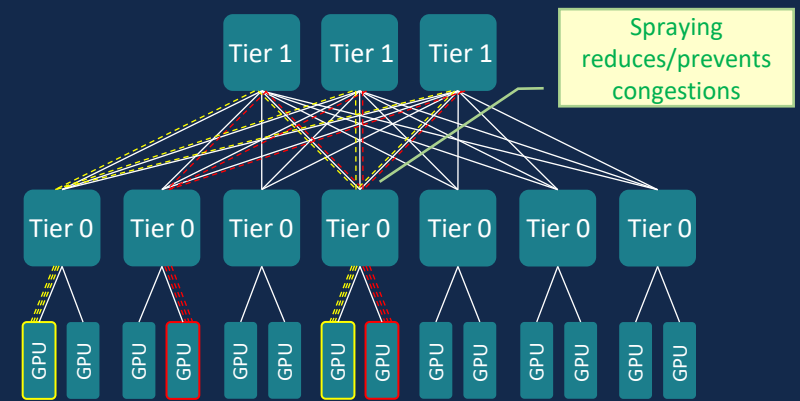
A software-defined acceleration layer that turns the interconnect from passive transport into an active, workload-aware participant in execution.

Without programmable spraying



Flows collide → hotspots, tail latency

With programmable spraying



Traffic spread evenly across equal-cost paths

Deploy

New congestion control, collectives, and routing
— without a respin

Tune

Tenant- and workload-specific optimization in the
data path

Evolve

Track UEC, RDMA extensions, and CXL services as
they emerge

THE POINT *Programmable hardware lets infrastructure innovation proceed at software velocity.*

Why programmability wins

Where a programmable infrastructure platform turns adaptability into measurable advantage — today, not next silicon cycle.



Feature velocity

New protocols and algorithms ship as updates, not board respins — months, not years.



Speed to solution

Prototype, validate, and deploy data-path changes against a live workload in one cycle.



Real-time telemetry

Line-rate congestion analytics and workload-aware visibility feed adaptive decisions.



Security as it arrives

Inline encryption, zero-trust east-west enforcement, and tenant isolation — added when needed.



Reconfigurability

The same platform absorbs the next standard, making adaptability a durable advantage.

The road ahead

- AI fabrics are evolving rapidly.
- Interconnects are becoming workload-aware.
- Innovation is outpacing traditional deployment cycles.
- **Tomorrow's AI factories will demand adaptive, intelligent networking.**

